

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Industry Problem Solving Task

Criteria	Understanding of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solution Design (2.5)	Use of Tools (2)	Analysis (1)	Total(10)
<i>Weightage</i>	20%	25%	25%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I S.Blessika(192525251) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:S.Blessika

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Annexure A

Q1 . Dataset: Supermarket Purchases

Customer Visits/Month Avg Spend (₹) Items Bought

C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Question:

Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. (BL3 – 7 Marks)

The screenshot shows the Weka Explorer interface with the SimpleKMeans clustering algorithm applied to the Supermarket Purchases dataset. The 'Cluster mode' is set to 'Use training set'. The 'Clusterer output' pane displays the following information:

- Number of iterations: 3
- Within cluster sum of squared errors: 3.1081506676744777
- Initial starting points (random):
- Cluster 0: C4,3,6000,6
- Cluster 1: C2,4,5000,8
- Missing values globally replaced with mean/mode
- Final cluster centroids:

Attribute	Full Data	Cluster# 0	Cluster# 1
	(5.0)	(2.0)	(3.0)
Customer	C1	C2	C1
Visits/Month	8	3.5	11
Avg Spend (?)	3380	5500	1966.6667
Items Bought	12.6	7	16.3333

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	2 (40%)
1	3 (60%)

Q2. Dataset: Credit Card Transactions

Transaction Amount (₹) Location Time (hrs)

T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11

Transaction Amount (₹) Location Time (hrs)

T4	7000	Mumbai	1
T5	180	Chennai	12

Question:

Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. (BL4 – 7 Marks)

The screenshot shows the Weka Explorer interface with the SimpleKMeans clustering algorithm applied to a dataset. The 'Clusterer' tab is active, and the 'SimpleKMeans' algorithm is selected. The 'Cluster mode' is set to 'Use training set'. The 'Clusterer output' pane displays the following information:

- Number of iterations: 3
- Within cluster sum of squared errors: 4.063311626771005
- Initial starting points (random):
- Cluster 0: T4,7000,Mumbai,1
- Cluster 1: T2,5000,Delhi,2
- Missing values globally replaced with mean/mode
- Final cluster centroids:
- Attribute Full Data Cluster#
- Transaction T1 T2 T1
- Amount (?) 2506 6000 176.6667
- Location Chennai Delhi Chennai
- Time (hrs) 7.2 1.5 11
- Time taken to build model (full training data) : 0 seconds
- === Model and evaluation on training set ===
- Clustered Instances
- 0 2 (40%)
- 1 3 (60%)

Q3. Dataset: Telecom Usage

User Call Duration (min) Data Usage (GB) Recharge (₹)

U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

Question:

Explain how clustering can be applied to identify churn-prone users and improve customer

retention strategies. (BL4 – 7 Marks)

The screenshot shows the Weka Explorer interface with the SimpleKMeans clustering algorithm applied to a dataset. The 'Clusterer' tab is active, displaying the command: `SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance" -R first-last" -I 500 -num-slots 1`.

Cluster mode:

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Clusterer output:

```

Number of iterations: 3
Within cluster sum of squared errors: 3.074911783724699

Initial starting points (random):
Cluster 0: U4,80,1,90
Cluster 1: U2,100,2,100

Missing values globally replaced with mean/mode

Final cluster centroids:
Attribute      Full Data      Cluster#
              (5.0)         (2.0)         (3.0)
=====
User           U1            U2            U1
Call Duration (min) 230          90      323.3333
Data Usage (GB)    3.9          1.5          5.5
Recharge (?)       164          95          210

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances
0      2 ( 40%)
1      3 ( 60%)
  
```

Result list (right-click for options):

- 14:13:46 - SimpleKMeans
- 14:25:00 - SimpleKMeans
- 14:31:11 - SimpleKMeans

Status: OK

Q4 . Dataset: E-commerce Behavior

User Product Views Clicks Purchases

U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

Question:

Suggest a clustering technique to group users and explain how it improves recommendation

systems. (BL3 – 7 Marks)

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance" -R first-last" -I 500 -num-slots 1

Cluster mode

☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Purchases ~
☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 14:13:46 - SimpleKMeans
- 14:25:00 - SimpleKMeans
- 14:31:11 - SimpleKMeans
- 14:38:09 - SimpleKMeans**

Cluster output

Number of iterations: 2
Within cluster sum of squared errors: 2.9243771626297583

Initial starting points (random):

Cluster 0: U3,60,25,6
Cluster 1: U1,50,20,5

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster#	0	1
	(4.0)		(1.0)	(3.0)
User	U1		U3	U1
Product Views	40		60	33.3333
Clicks	15.75		25	12.6667
Purchases	3.5		6	2.6667

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	1 (25%)
1	3 (75%)

Status OK Log x 0

Q5. Dataset: Traffic Data

Location Vehicle Count Avg Speed (km/h) Delay (min)

L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Question:

Design a clustering-based solution to identify traffic congestion zones and explain how it

helps in traffic management. (BL4 – 7 Marks)

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1

Cluster mode

☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Delay (min) v
☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 14:13:46 - SimpleKMeans
- 14:25:00 - SimpleKMeans
- 14:31:11 - SimpleKMeans
- 14:38:09 - SimpleKMeans
- 14:40:08 - SimpleKMeans**

Clusterer output

Number of iterations: 3
Within cluster sum of squared errors: 3.113782053684652

Initial starting points (random):

Cluster 0: L4,600,10,30
Cluster 1: L2,500,15,25

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data (5.0)	Cluster#	
		0 (2.0)	1 (3.0)
Location	L1	L2	L1
Vehicle Count	346	550	210
Avg Speed (km/h)	23	12.5	30
Delay (min)	17.2	27.5	10.3333

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	2 (40%)
1	3 (60%)

Status
OK

Log x 0